

Notion of a Set

A set is a collection of objects. One would immediately want to ask, what do you mean by “collection”? While we do not give a definition of a collection, we can assert that given an object, we can decide if it belongs to a collection or not. For example, a collection of apples is intuitively well-defined, in the sense that given a fruit, we can, in almost all cases, decide if it is an apple or not, and so if it belongs to the collection of apples or not.

Since we are talking about a set in an abstract way, we will want to give it a notation, or a symbolic representation. Usually sets are denoted by upper case letters, such as A .

If A is a set, we will indicate that some object, denoted by x , belongs to the set A by writing

$$x \in A.$$

Similarly the notation

$$x \notin A$$

stands for **x not in the set A** . Members of a set are called **elements of the set**. So we say that $x \notin A$ means x is not an element of A .

Describing a set

Depending on the nature of the set, a set can be described (or defined) in a few different ways.

1. Enumeration

Explicitly list the elements of a set, using curly brackets

$$A = \{1, 2, 3, 4, 5\}$$

$B = \{1, 2, 3, 4, \dots, 10\}$, where the three dots, called *ellipsis*, stands for the elements that, quite clearly, are the remaining integers 5 through 9. We ellipses when it is clear what elements are included but not listed explicitly.

$C = \{1, 2, 3, 4, \dots\}$ is an *infinite* set of all positive integers starting from 1, increasing by 1.

$D = \{\text{Corolla, Camry, Rav4, } \dots\}$ is a set of all car models made by Toyota. You may not even know all members of this set, but in principle, given a car, x , you can decide if $x \in D$ or $x \notin D$.

Sets that are frequently encountered are usually given symbols that are reserved for them alone. For example, since we will be referring to the positive integers throughout this book, we will use the symbol $\mathbb{P}$ instead of writing $\{1, 2, 3, \dots\}$.

A few of the other sets of numbers that we will use frequently are:

- $\mathbb{P}$: the positive integers, $\{1, 2, 3, 4, \dots\}$
- $\mathbb{N}$: the natural numbers, $\{0, 1, 2, 3, \dots\}$
- $\mathbb{Z}$: the integers, $\{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$
- $\mathbb{Q}$: the rational numbers
- $\mathbb{R}$: the real numbers
- $\mathbb{C}$: the complex numbers

Notice the difference between $\mathbb{P}$ and $\mathbb{N}$. Some people call $\{1, 2, 3, 4, \dots\}$ the set of *natural numbers*, but we include 0 in the the set we call *natural numbers*.

2. Set-Builder Notation.

Another way of describing sets is to use set-builder notation. For example, we could define a set of all even positive numbers:

$E = \{x \in \mathbb{P} \mid x \text{ is even}\}$. This is read as a set consisting of all elements, x in the set of positive numbers such that x is even (divisible by 2). The $\mid$ symbol is read as “such that”.

$\mathbb{Q} = \{\frac{a}{b} \mid a, b \in \mathbb{Z}, b \neq 0\}$ is the set of rational numbers. In this notation,

- $\frac{a}{b}$ indicates that a typical element of the set is a “fraction.”
- The vertical line, $\mid$, is read “such that” or “where,” and is used inter-changeably with a colon.
- $a, b \in \mathbb{Z}$ is an abbreviated way of saying a and b are integers.
- Commas is read as “and.”

Order of elements and duplicates in sets

In a set, A , the *order of elements do not matter*. So the set $\{1, 2, 3\}$ is the same as the set $\{3, 1, 2\}$.

A set doesn’t have duplicates. If there are repeating elements, they are considered one and the same.

A set shouldn’t have duplicates. The set $\{a, b, c\}$ is the same as the set $\{a, b, c, a\}$, which is the same as $\{a, b, b, c\}$. You only write the unique elements in the set representation.

Finite and infinite sets

As the name implies, finite sets are defined as those with a finite number of elements in them. An infinite set has an infinite number of elements in them.

Cadinality of sets

For the time being, we define *cadinality of a finite set* as the number of elements in the set, denoted by $|A|$.

There are different types of cardinalities defined for infinite set, but we will not talk about them now.

If $A = \{1, 2, 10, 8, 6, 5\}$, then $|A| = 6$.

Subsets

Let A be some set. Another set, B, is considered a subset of A if every element of B is also an element of A. We use the notation $A \subseteq B$ to say **A is a subset of B**.

Notice that according to this definition, A is a subset of itself! Think about this. **So it is always true that $A \subseteq A$.**

If $A = \{1, 2, 3, 4, 5\}$, and $B = \{3, 4, 5\}$, then $B \subseteq A$. It is, however, not true that $A \subseteq B$, since there is an element in A which is not in B.

Supersets

If $A \subseteq B$ then B is called a **superset** of A.

Set equality

Two sets A and B are defined to be equal, *if and only if* every element of A is in B, and, every element of B is in A. In other words, For two sets A and B, $A=B$ if and only if $A \subseteq B$ and $B \subseteq A$. So two sets are equal if (and only if) they have the same elements.

If and only if

The phrase *if and only if* needs some thought. Consider the statements “I will run if it rains” and “I will run if and only if it rains”. Can you see how these two phrases are not the same in their meanings?

Empty set

Sometimes we need to refer to a set that has no elements, or an empty set. We use the notation $\emptyset$ or $\varnothing$ or $\{\}$ to denote an empty set. **The cardinality of $\emptyset$ is zero.** It is a somewhat special set, because it is considered a subset of every set. In other words, $\emptyset \subseteq A$ for any set A .

Proper and improper subsets

We saw before that for any set A , $A \subseteq A$. In this case we call A an **improper** subset of itself. All other subsets of A are called **proper** subsets of A . The empty set, $\emptyset$, is considered a proper subset of any nonempty set A . The empty set, $\emptyset$, is considered an improper subset of itself.

Note that if A is a proper subset of B , then all elements of A are in B , but there is at least one element in B that is not in A .

Set operations

Section 1.2.1

(a) *Intersection* of two sets $A \cap B$

$$A \cap B = \{x : x \in A \text{ and } x \in B\}$$

(b) *Union* of two sets $A \cup B$

$$A \cup B = \{x : x \in A \text{ or } x \in B\}$$

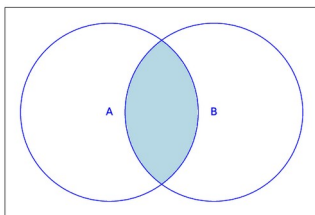
(c) A and B are *disjoint sets* $A \cap B = \emptyset$

(d) *The Universe, U* Superset of all sets.

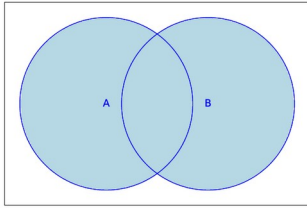
Section 1.2.2 Set Operations and their Venn Diagrams

When working with sets, as in other branches of mathematics, it is often quite useful to be able to draw a picture or diagram of the situation under consideration. A diagram of a set is called a Venn diagram. The universal set U is represented by the interior of a rectangle and the sets by disks inside the rectangle.

Venn diagram of the intersection and union of two sets.



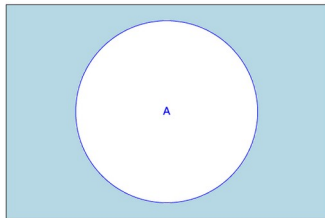
Intersection of two sets A and B



Union of two sets A and B

Complement of a set, A. Elements of the Universe not in A. Notation: A^c

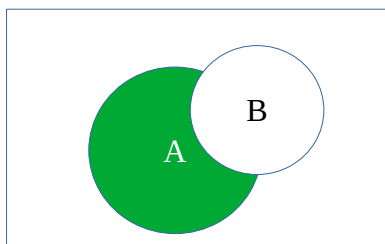
$A^c = \{x \in U | x \notin A\}$. As a consequence of the definitions, $\emptyset^c = U$ and $U^c = \emptyset$



The blue area is the complement of A

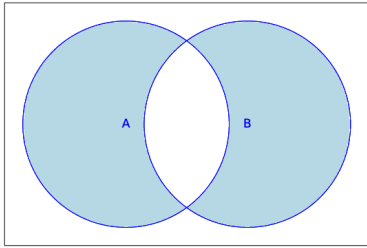
The subtraction operation $A - B$ between two sets A and B is defined as

$A - B = \{x \in A | x \notin B\}$. $A - B$ is the elements of A which are not in B. $B - A$ will be elements in B not in A.



The green represents $A - B$. This is elements that are in A but not in B

Let A and B be sets. The symmetric difference of A and B (denoted by $A \oplus B$) is the set of all elements that are in A and B but not in both. That is, $A \oplus B = (A \cup B) - (A \cap B)$



The blue area represents $A \oplus B$